Performance Metrics & KNN for Regression

As we know that for a classification problem, the performance metric is Accuracy.

Accuracy → Out of total predictions, how many are correct predictions.

It does not tells us the performance of the model on individual class.

In such cases, we go for some other metrics

① Precision
② Recall

Suppose there are two classes → C_1 = +ve (1) , C_2 = -ve (0)

Precision → Out of all predicted +ve, how many are correctly classified by the model.

Recall → Out of all actual +ve, how many are correctly classified by the model.

Ex-

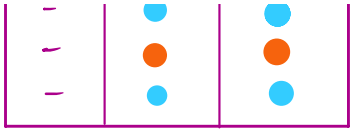
X	Y	Y_P
–	●	●
–	●	●
–	●	●
–	●	●
–	●	●
–	●	●
–	●	●
–	●	●

● → -ve
● → +ve

$Acc = \frac{5}{8}$

Precision => Pred. +ve = 5, Corr. +ve = 3 ⇝ $\frac{3}{5} = 0.6$

Recall => Actual +ve = 4 ⇝ $\frac{3}{4} = 0.75$



Recall ⇒ Actual +ve = 4 ⇝ $\frac{3}{4}$ = 0.75
Corr. +ve = 3

NOTE:- The range of accuracy, precision, recall is from 0 to 1
The more the score is closer to 1, the better are the results

★ KNN for Regression :-

When data is linear then → linear regression
" data is non-linear then → KNN

Steps :-

1. Take a query point & all the data points
2. Calculate the distance blw the query point & all data points
3. Choose a value of K
4. Select K Nearest Neighbors
5. Calculate the average of the top K neighbors
6. This avg value is the predicted value for the query point.

★ Disadvantages of KNN :-

1. KNN does not perform well in case of spiral data.



2. Inference time of KNN is very high.

Ex- Training → 8 lac
Test → 2 lac
} KNN fails with huge data